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SOLVED NUMERICALS – JAN/FEB 2023 (CSE STREAM)

1(C) The acceptance angle of an optical fiber is 30° when kept in an air find the angle of acceptance when it is in a medium of refractive index 1.33

(Dec 2017 | 17S) (CSE Stream, JAN/FEB – 2023 | 22S)

Data: $Refractive\ index\ of\ core\ , n_0' = 1.33$

$Angle\ of\ acceptance\ , \theta = 30°$

To find: Angle of acceptance while the fiber is in the refracting medium, $\theta_0' = 1.33$

Solution: we have the equation, $sin\theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0'}$ .....(1)

When the surrounding medium is air, then $n_0 = 1 and\ \theta_0 = 30°$

$$sin30° = \sqrt{(n_1^2 - n_2^2)}$$

$$0.5 = \sqrt{(n_1^2 - n_2^2)} \text{.....(2)}$$

When the surrounding medium is of the refractive index 1.33, let θ_0' be the acceptance angle.

Now eqn(1) becomes, $\sin\theta_0' = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0'} = \frac{\sqrt{(n_1^2 - n_2^2)}}{1.33}$

$$1.33 \sin\theta_0' = \sqrt{(n_1^2 - n_2^2)} \text{ .....(3)}$$

$\therefore from\ equation(2) and\ (3),$ $1.33 \sin\theta_0' = 0.5$

$$\theta_0' = sin^{-1}\left(\frac{0.5}{1.33}\right) = 22°$$

$\therefore$ ***The Acceptance angle when the fiber is in the medium of refractive index*** **1.33** ***is*** **22°**

2(C) In a diffraction grating experiment, the laser light undergoes second order diffraction for diffraction angle 1.48°. The grating constant is 5.08×10^{-5}m and the distance between the grating and the source is 80cm. Find the wavelength of the laser light. **(CSE Stream, JAN/FEB – 2023 | 22S)**

Given: $m = 2$, $d = 5.08\times10^{-5}$ m, $f = 80$cm & $\theta = 1.48°$.

Wavelength $\lambda = \frac{d\sin\theta}{m}$

$$\therefore \lambda = \frac{5.08\times10^{-5}\times\sin(1.48°)}{2}$$

$$\lambda = \frac{5.08\times0.026\times10^{-5}}{2} = 0.0656\times10^{-5} = 656\times10^{-9}\,m$$

i.e., $\lambda = \mathbf{656nm}$

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3(C) Compute the de-Broglie wavelength associated with neutron of mass 1.674×10^{-27} kg moving with one tenth part of the velocity of light.

(DEC 2018 – 17S) (CSE Stream, JAN/FEB – 2023 | 22S)

Data: Mass of the neutron: 1.674×10^{-27} kg,

Velocity of light, $c = 3 \times 10^8\, m/s$

Velocity of neutron $v = \frac{1}{10} \times c = 3 \times 10^7 m/s$

To Find: De Broglie wavelength, $\lambda = ?$

Solution: De Broglie wavelength, $\lambda = \frac{h}{mv}$

$$\lambda = \frac{6.6 \times 10^{-34}}{(1.074 \times 10^{-27})(3 \times 10^7)} = 1.314 \times 10^{-14}\, m$$

$$\therefore \boldsymbol{De - Broglie\ wavelength\ associated\ of\ the\ neutron\ is\ 1.314 \times 10^{-14}\ m}$$

4(C) An electron has a speed of 100m/s. The inherent uncertainity in its measurement is 0.005%. Calculate the corresponding uncertainity in the measurement of the position.

(CSE Stream, JAN/FEB – 2023 | 22S)

Solution: $\Delta v = \mathbf{100} \times \frac{\mathbf{0.005}}{\mathbf{100}} = \mathbf{0.005}\ \boldsymbol{m/s}$

$$\Delta x \geq \frac{h}{4\pi m \Delta v} = \frac{6.625 \times 10^{-34}}{4 \times 3.14 \times 9.11 \times 10^{-31} \times 0.005} = 0.0115\, m$$

Uncertainty in the measurement of the position, $\Delta x = 0.0115\, m$

5(C) Show that Hadamard gate is unitary.

Solution: $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$H.H^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H.H^\dagger = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$H.H^\dagger = \begin{pmatrix} \frac{1}{2}+\frac{1}{2} & \frac{1}{2}-\frac{1}{2} \\ \frac{1}{2}-\frac{1}{2} & \frac{1}{2}+\frac{1}{2} \end{pmatrix}$$

$$H.H^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1, Hence\ Hadamard\ gate\ is\ Unitary.$$

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6(C) *A linear operator 'X' operates such that* $X|0\rangle = |1\rangle$ *and* $X|1\rangle = |0\rangle$. Find the matrix representation of 'X'. **(CSE Stream, JAN/FEB – 2023 | 22S)**

Solution: $\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}$, $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ *and* $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$Given\ X|0\rangle = |1\rangle \qquad X|0\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get, $x_{11} = 0$ & $x_{21} = 1$

$Given\ X|1\rangle = |0\rangle \qquad X|1\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get, $x_{12} = 1$ & $x_{22} = 0$

$Therefore, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

7(C) Find the temperature of which there is 1% probability that a state with an energy 0.5eV above the Fermi energy is occupied.
(Jan 2019 -18S | July 2022 – 21S) (CSE Stream, JAN/FEB – 2023 | 22S)

Data: $\boldsymbol{Probability, f(E) = 1\% = 0.01}$
Energy above E_F *i.e.*, $(E - E_F) = 0.5eV = 0.5 \times 1.602 \times 10^{-19} J$
To find : $Temperature\ at\ which\ the\ above\ data\ holds\ good, T = ?$
Solution: we have $f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$

$\frac{E-E_F}{kT} = \frac{0.5 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times T} = \frac{5797}{T}$

Substitute the above value, we have $0.01 = \frac{1}{e^{\frac{5797}{T}} + 1}$

$e^{\frac{5797}{T}} = \frac{1}{0.01} - 1 = 99$

Taking natural log on both sides we have, $\frac{5797}{T} = In99$

$(\therefore In\ 99 = 4.595)$

$\therefore \boldsymbol{T = \frac{5797}{4.595}}$

$\boldsymbol{T = 1261.1K}$

$\therefore \boldsymbol{Temperature\ at\ which\ the\ above\ data\ holds\ good, T = 1261.1K}$

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8(C). The critical temperature of Nb is 9.15K. At zero kelvin, the critical field is 0.196T. Calculate the critical field at 8K?

$$H_C = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$H_C = 0.196\left[1 - \left(\frac{8}{9.15}\right)^2\right]$$

$$H_C = 0.046\,T$$

9(C) A slowing in object in an animation has a first frame distance 0.5m and first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.

(CSE Stream, JAN/FEB – 2023 | 22S)

$$base\ distance = \frac{distance\ between\ any\ two\ adjacent\ frames}{2}$$

$$base\ distance = \frac{0.5 - 0.35}{2} = 0.075m$$

To find number of frames

$$No.\ of\ frames = \frac{total\ distance}{base\ distance}$$

$$No.\ of\ frames = \frac{0.5}{0.075} = 6.66 \approx 7\ frames$$

10. (C) In an optical fiber experiment, the laser light propagating through optical fiber cable of length 1.5 m, made spot diameter of 8 mm on the screen. The distance between the end of the optical fiber and the screen is 3.4 cm. Calculate the angle of acceptance and numerical aperture.

(CSE Stream, JAN/FEB – 2023 | 22S)

Given: $D = 8$ mm $= 0.8$ cm & $l = 3.4$ cm.

Angle of acceptance is given by $\tan\theta_o = \frac{D}{2l}$

$$\therefore \theta_o = \tan^{-1}\left(\frac{D}{2l}\right) = \tan^{-1}\left(\frac{0.8}{2\times 3.4}\right) = \tan^{-1}(0.118) = \mathbf{6.71^o}$$

Numerical aperture $NA = \sin\theta_o$

$NA = \sin(6.71^o) = \mathbf{0.117}$

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